

Tufte-Dark Slide Catalogue

Each layout, and when to reach for it

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Image

**Most of the world's data
has **never** been looked at.**

1

Content layouts

text, math, and boxes

Margin note: a citation or side caveat

Lloyd's algorithm alternates assignment and update until the partition stops changing. Each step cannot increase $\mathcal{J}$, so it converges, though only to a **local** minimum.

The centroid update is just the per-cluster mean: the choice that minimises that cluster's contribution to the objective.

Lloyd, 1957; published 1982.

The margin holds the aside, the citation, the small caveat: present, but out of the main line of thought.

Equation: the 2/3 + 1/3 grid

$$\mathcal{J}(C, \mu) = \sum_{k=1}^K \sum_{x \in C_k} \|x - \mu_k\|^2$$

The summed squared distance from every point to **its own centroid**. Driving it down is precisely what we mean by a coherent cluster.

Definition, with a reference in the margin

Definition: Centroid

The centroid μ_k of cluster C_k is the mean of its members: the point that minimises the summed squared distance within the cluster.

Exactly the per-cluster term of $\mathcal{J}$.

Lloyd (1982), for the original quantisation argument.

Introducing a key result

Density-based clustering recovers groups of **arbitrary** shape and marks sparse points as noise, without fixing the number of clusters in advance.

Ester et al. (1996)

A density-based algorithm for discovering clusters. *Proc. KDD*.

$$x \text{ is a core point} \iff |N_\epsilon(x)| \geq \text{minPts}$$

Status blocks: each one signals a status

Definition or statement

Standard block, grey title bar. The thing to remember: a definition, a property, a result.

Worked example

Example block, second tone. A concrete instance that makes the abstract statement tangible.

The one caveat

Alert block, red. Reserved for the warning you must not miss; this red bar is the slide's single accent.

Two columns: a side-by-side comparison

Lloyd's algorithm

- Assign each x to nearest μ_k .
- Recompute each μ_k .
- Repeat until assignments settle.

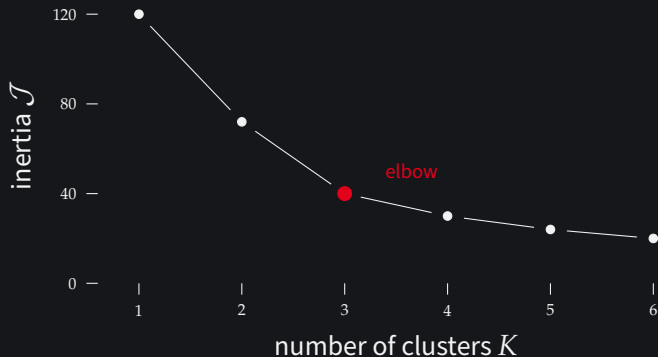
Caveats

- Converges to a **local** minimum.
- Sensitive to initialisation.
- Assumes roughly spherical clusters.

2

Figures and images drawn or dropped in

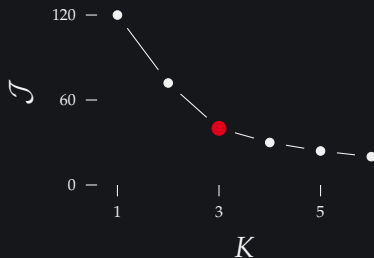
Figure: a plot on the 2/3 + 1/3 grid



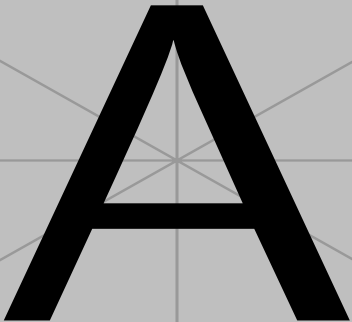
The artefact lives in the main two-thirds, its reading in the right third. The **elbow** at $K = 3$ is where extra clusters stop paying for themselves.

Equation and figure, three columns

$$\mathcal{J} = \sum_{k=1}^K \sum_{x \in C_k} \|x - \mu_k\|^2$$



The objective falls fastest early; the **elbow** marks where extra clusters stop paying off.



Latent space of the **learned** encoder

Full-height side image, text beside it

A portrait image can bleed the **full slide height** on one edge, with the text held in a **width-matched** column beside it.

- `\sideimage[0.30]{img}` sets the image, right edge.
- `\sideimage*` puts it on the left instead.
- wrap the prose in the `sidebody` environment.

Keep the text concise: the image is the hero, this column is the caption.

16×9

(Original size: 180×320 bp)

GIF: an animation, filling, with a margin note

GIF sample not rendered.

First build needs `-shell-escape` and `ffmpeg`; recompile once with both.

Drop `clip.gif` in `gifs/` and write `\gif{clip}`: the theme splits and caches the frames for you.

It plays in Acrobat and `pdftpc`; elsewhere the **poster** frame stands in.

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Tables, code and algorithms

tables, code, and steps

Table: full width, Tufte, caption in the margin

method	needs K	cluster shape
<i>k</i> -means	yes	spherical
Ward linkage	no	compact
DBSCAN	no	arbitrary

One axis of choice: must you fix K up front? *k*-means does; the others infer it from the data.

Pseudocode, with a margin note

Algorithm 1: Lloyd's algorithm for k -means

Input: data X , cluster count K

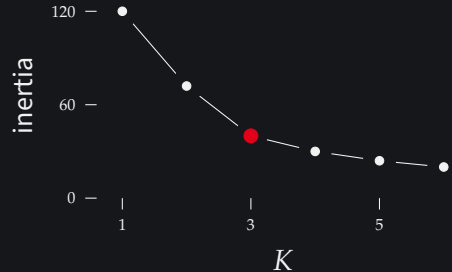
Output: assignments C , centroids μ

- 1 initialise $\mu_1, \dots, \mu_K$
 - 2 **repeat**
 - 3 assign each x to its nearest centroid μ_k
 - 4 recompute each μ_k as the mean of its members
 - 5 **until** *assignments stop changing*
-

The two loop lines are the whole idea: **assign**, then update. Everything else is bookkeeping.

Code and its output, side by side

```
from sklearn.cluster import KMeans
inertias = [KMeans(k).fit(X).inertia_
             for k in range(1, 7)]
```

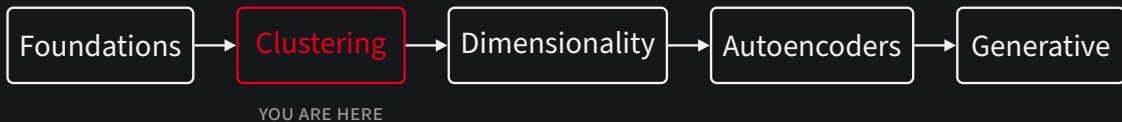


4

Teaching and flow

openers and recaps

Roadmap: where we are in the course



Five weeks in. The map keeps the whole arc visible between sessions.

Recap: three takeaways

Objective first

The objective defines the clustering, not the algorithm.

Shape is assumed

k -means expects spherical, comparable-scale clusters.

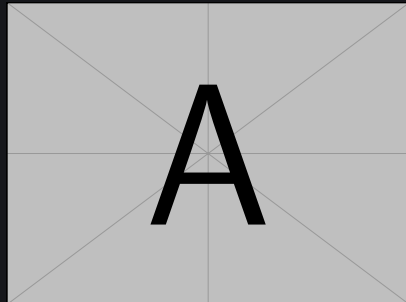
The elbow hints

It **suggests** K ; it does not prove it.

Question: a prompt, with room for a visual

Think for a minute

How would you choose K **without ever seeing the labels?**



Key number: one result, large

87%

The first three clusters already capture most of the structure. Adding more explains only **diminishing** extra variance: past the elbow, the model fits noise, not signal.

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Closing

recap and thanks

Close: what you can now do

Find groups

Recover structure with no labels at all.

Judge how many

Read the elbow, but do not **trust** it blindly.

Read meaning

Standardise first, then interpret each cluster.

Thanks for having me

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